

**MASH**

MISHMASH OPENING CONFERENCE · MACHINES SESSION

# Building a Norwegian Ecosystem for AI and Creativity

Kilden, Kristiansand · 14 September 2026





### THREE QUESTIONS I GET

**1** "How do I get access to train a model on Sigma2?"

Which tools *can* we use?

**2** "Can I get funding for Claude Code?"

Which tools *should* we (not) use?

**3** "How do I send an avatar of myself from Bergen to Trondheim in 10 milliseconds?"

Which tools do we need to *develop*?



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**Rent or run**





## Rent a service

ChatGPT · Claude · Gemini · Suno · Runway · Midjourney

- + the best models, today
- + zero setup, pay per use
- + works on any device

- no weights, no reproducibility
- your data leave the room
- terms and prices change
- same for everyone, no yours

control: theirs

## The middle road

open models,  
rented GPUs  
(HF, Mistral,  
EU providers)

## Run it yourself

open models on hardware you or your institution control

- + full control, reproducible
- + private: nothing leaves
- + can be trained on your material

- one step behind the frontier
- you carry the setup
- you pay the energy bill
- needs hardware and time

control: yours





# Six questions to ask of any tool

## Which model?

Name and version. "AI" is not an answer.

## Where does it run?

Your machine, your institution, an EU data centre, the US.

## Where do my data go?

Prompt, files, recordings. Stored? Trained on?

## Who owns the output?

And can you sell a work made with it?

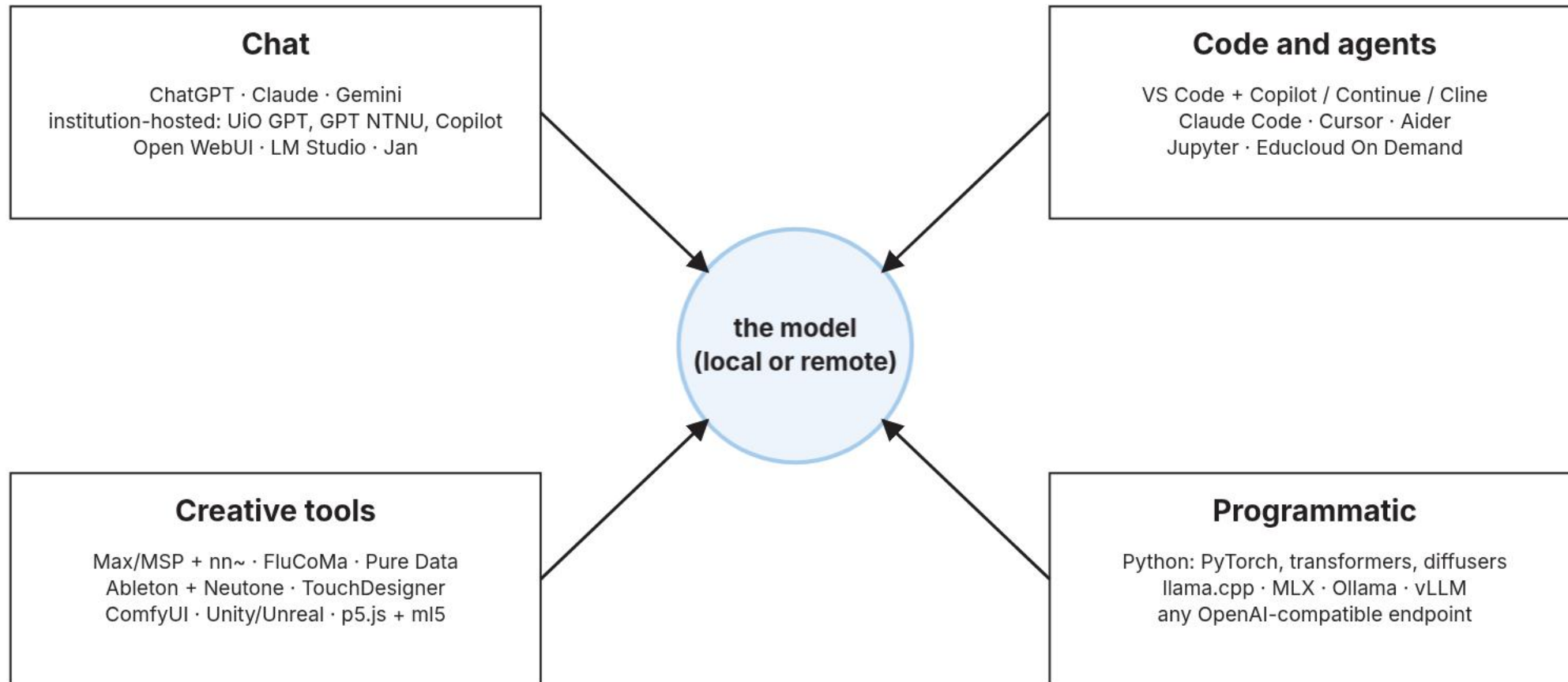
## Same result next year?

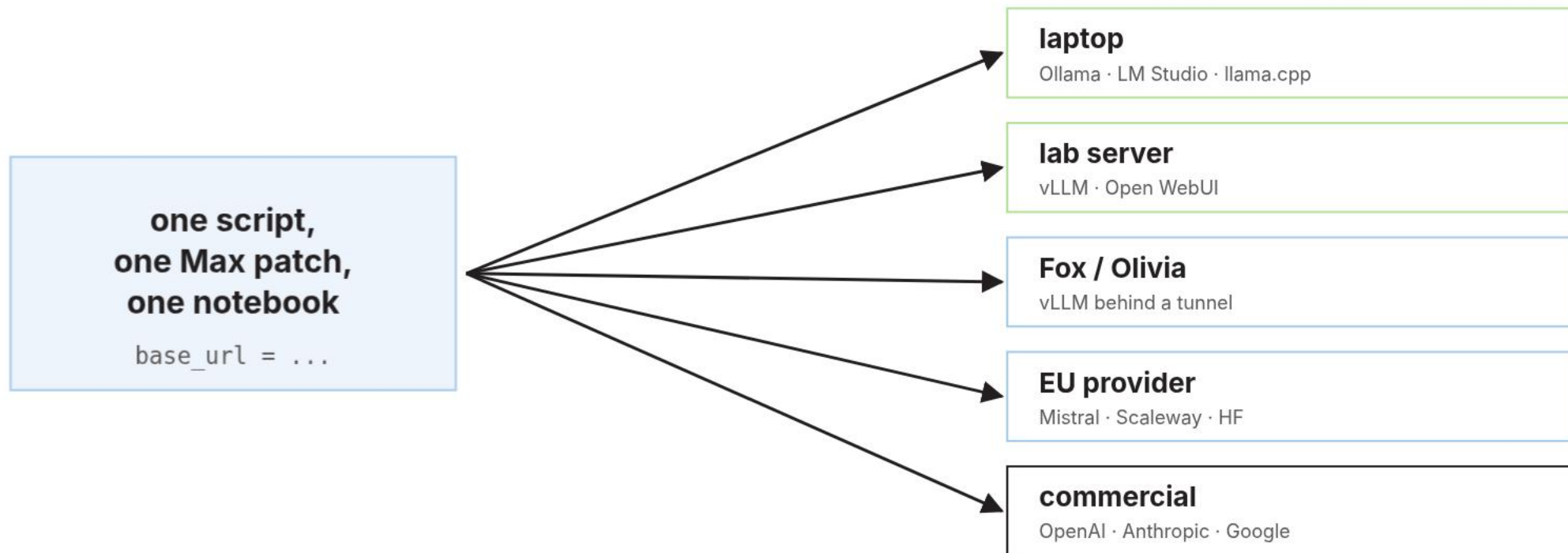
Reproducibility for research; continuity for a repertoire piece.

## What does it cost?

Per month, per hour of GPU, per kilowatt-hour.



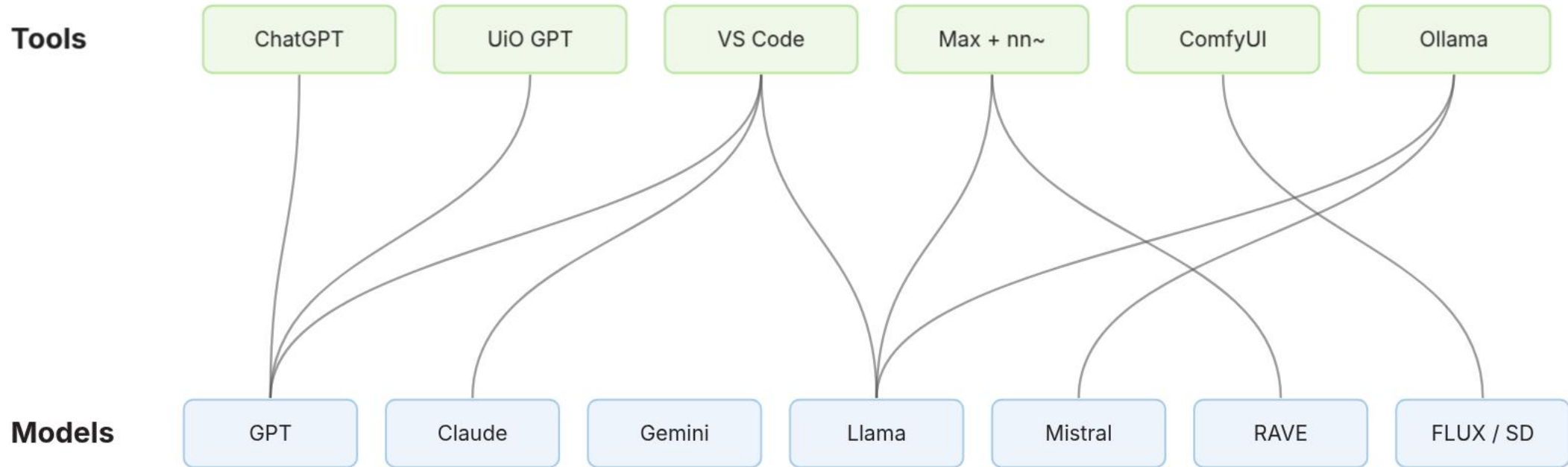




the same OpenAI-compatible API on all of them: change one line, keep the workflow



# Tool is not model





## Language

Llama · Mistral · Gemma · Qwen ·  
DeepSeek · Phi

Norwegian: NorMistral (UiO) · NorLLM  
(NorwAI/NTNU) · NB-GPT

## Speech

Whisper · faster-whisper · Piper · XTTS  
· Kokoro

Norwegian: NB-Whisper (National  
Library)

## Music and audio

RAVE (IRCAM) · MusicGen / AudioCraft  
· Stable Audio Open · ACE-Step · YuE

**Norwegian: none**

## Image

Stable Diffusion · SDXL · FLUX

**Norwegian: none**

## Video

Wan · HunyuanVideo · LTX

**Norwegian: none**

## Vision, motion, multimodal

CLIP · SAM · LLaVA · Qwen-VL · MDM ·  
MotionGPT

**Norwegian: none**





# Licences in three colours

## Anything goes

Apache 2.0 · MIT

- Mistral, Mixtral · Qwen3 · DeepSeek · Phi
- NorMistral (UiO) · NB-Whisper
- Whisper · RAVE · CLIP · SAM
- FLUX.1 schnell · Wan 2.1 · YuE · ACE-Step
- use it, sell the work, ship the product

## Conditions

Community licences and terms of use

- Llama: "Built with Llama", 700M-user cap, derivatives named Llama
- Gemma: prohibited-use policy passed on
- Stability community licence (SD3.5, Stable Audio Open): free under \$1M revenue
- CreativeML OpenRAIL-M (SD 1.5, SDXL): use restrictions travel with the weights
- NorwAI NorLLM: Nordic organisations

## Non-commercial

Research only, CC-BY-NC

- **FLUX.1 dev: non-commercial licence for the model**
- **MusicGen: code MIT, weights CC-BY-NC 4.0**
- **many research checkpoints on Hugging Face**
- **fine for a paper; a commissioned work is commercial**



# Licence compatibility: what you can do with what

| Base model licence                                                                 | Research paper | Commissioned work, sold                        | Product or service                  | Release your fine-tune                 | Mix with a permissive model                | Train a new model on its outputs |
|------------------------------------------------------------------------------------|----------------|------------------------------------------------|-------------------------------------|----------------------------------------|--------------------------------------------|----------------------------------|
| <b>Apache 2.0 / MIT</b><br>Mistral, Qwen3, NorMistral, Whisper, RAVE, FLUX schnell | •              | •                                              | •                                   | • any licence, keep the notice         | •                                          | •                                |
| <b>Llama community</b><br>Llama 3, Llama 4                                         | •              | • "Built with Llama"                           | 🔒 under 700M users                  | 🔒 same licence, name starts with Llama | 🔒 the mix inherits Llama terms             | 🔒 allowed, new model named Llama |
| <b>Gemma terms of use</b><br>Gemma 2, 3                                            | •              | •                                              | 🔒 prohibited-use policy             | 🔒 terms passed on                      | 🔒 the mix inherits the terms               | 🔒 check the terms                |
| <b>Stability community</b><br>SD3.5, Stable Audio Open                             | •              | 🔒 free under \$1M revenue                      | 🔒 same cap, else enterprise licence | 🔒 same licence                         | 🔒 the mix inherits the cap                 | 🔒 check the terms                |
| <b>OpenRAIL-M</b><br>SD 1.5, SDXL                                                  | •              | •                                              | 🔒 use restrictions apply            | 🔒 restrictions passed on               | 🔒 restrictions travel with it              | 🔒                                |
| <b>Non-commercial</b><br>FLUX.1 dev, MusicGen weights, CC-BY-NC                    | •              | 🔒 read it: outputs sometimes yes, the model no | ○                                   | 🔒 non-commercial only                  | ○ the whole release becomes non-commercial | ○ usually forbidden              |





# Where to find them, how to trust them

## Find

- Hugging Face: weights, model cards, datasets
- Ollama and LM Studio libraries: one-click local
- IRCAM Forum: RAVE and nn~
- the paper's GitHub, for the newest

## Trust

- model card: training data disclosed?
- evaluation: on what, by whom?
- reproducible: pinned version, checksum
- energy: training and inference cost stated?



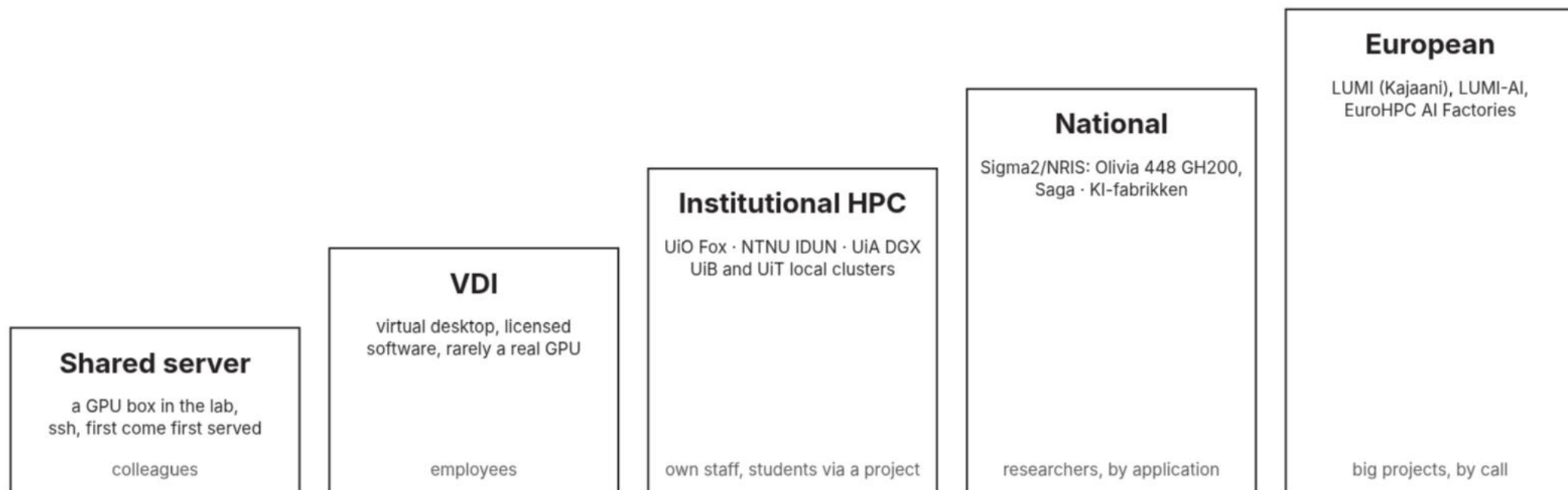


# Where the compute is



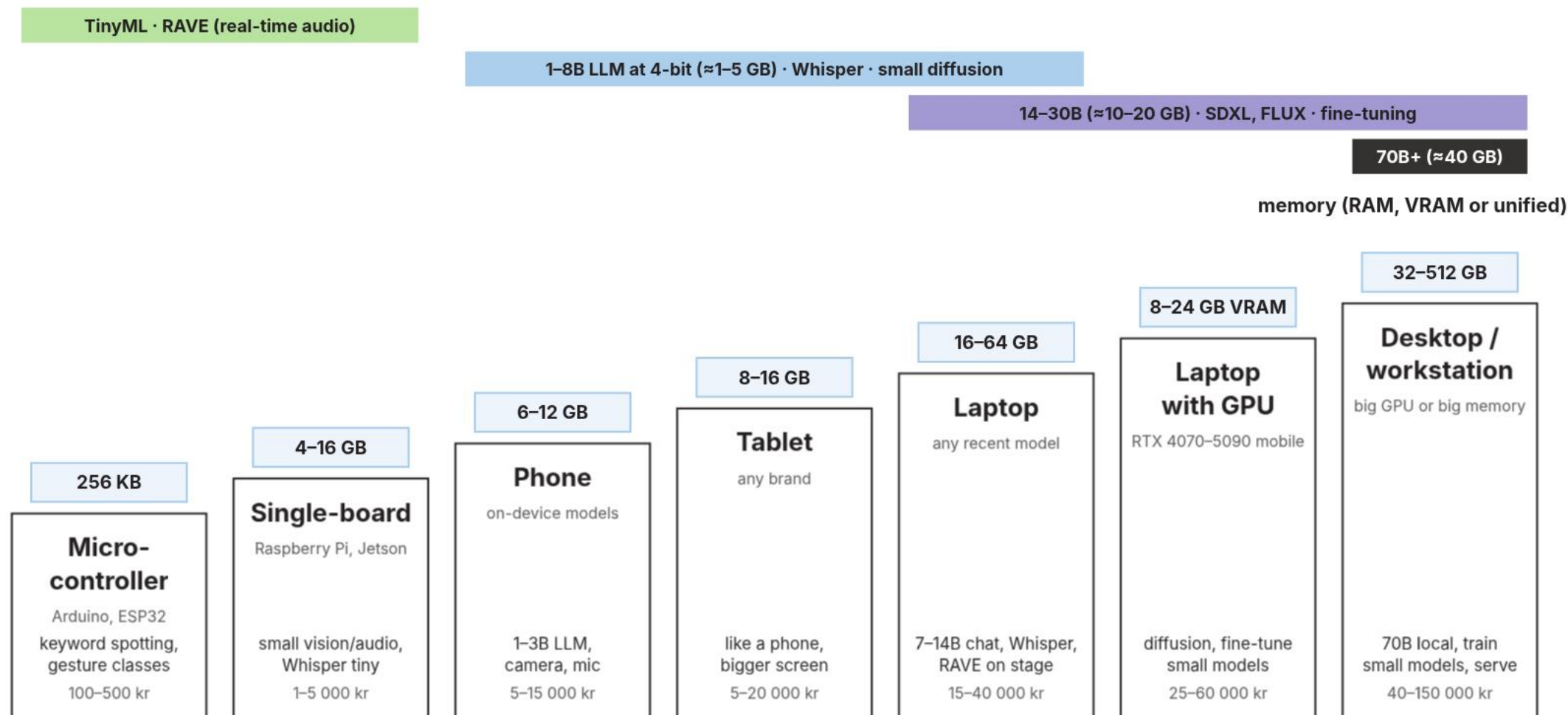


more compute · more people can share it · more data next to it



more waiting · more paperwork · less interactive · batch, not a rehearsal







**parameters × bytes per parameter ≈ memory**

|             |        |                 |                                           |
|-------------|--------|-----------------|-------------------------------------------|
| <b>7–8B</b> | 4-bit  | <b>≈ 5 GB</b>   | any recent laptop or phone                |
| <b>30B</b>  | 4-bit  | <b>≈ 20 GB</b>  | laptop with 32 GB unified, or a 24 GB GPU |
| <b>70B</b>  | 4-bit  | <b>≈ 40 GB</b>  | large unified memory, or two big GPUs     |
| <b>70B</b>  | 16-bit | <b>≈ 140 GB</b> | a server, or Fox                          |



# What each rung is good for

## Microcontroller

- keyword spotting, gesture classes
- sensor fusion, milliseconds
- no LLM, ever
- wearables, instruments

## Pi / Jetson

- small vision and audio models
- Whisper tiny, YOLO
- an embodied agent in a box
- installations that run for a year

## Phone / tablet

- 1–3B on-device LLMs
- camera, mic, sensors
- closed platforms, app stores
- the device every pupil has

## Laptop

- 7–14B chat, offline
- Whisper transcription
- RAVE and nn~ on stage
- the studio in a bag

## Laptop with GPU

- Stable Diffusion, FLUX
- fine-tune small models (LoRA)
- real-time everything
- hot, loud, one hour of battery

## Workstation

- 70B local, video generation
- train RAVE overnight
- serve a whole studio or class
- needs a room and a power bill

## School tablet or laptop

- managed device, no GPU
- Feide login, pupils' data
- everything must be hosted
- the most constrained rung

## Rule of thumb

- memory decides what fits
- compute decides how fast
- prototype on the smallest rung that works





# How you actually get on

## Institutional clusters

- UiO Fox: Educloud project, PI adds collaborators
- NTNU IDUN: supervisor requests access
- UiA: DGX servers at CAIR
- UiB and UiT: local HPC via the IT division

## NRIS / Olivia

- project application, CPU and GPU hours
- two rounds a year
- any Norwegian research institution
- data storage next to compute

## KI-fabrikken

- Sigma2, part of the LUMI AI Factory network
- research, business, public sector
- expertise and support, not only GPUs
- artists and cultural institutions: unclear

## LUMI / EuroHPC

- through NRIS or EuroHPC calls
- AMD GPUs: check your software
- large allocations, long lead times

## NAIC (paused)

- the low-threshold national GPU service
- funding ended 31 March 2026
- "no-cost hibernation" since April
- successor being procured by Sigma2

## Sensitive data

- TSD at UiO, SAFE at UiB, HUNT Cloud at NTNU
- GPU capacity inside is limited
- health and pupils' data live here or offline





# Institutions side by side · to be filled in

| Institution                                       | Own GPU cluster                     | Hosted chat / LLM service | Sensitive-data environment | Students get GPUs? | Freelancers / partners get in? |
|---------------------------------------------------|-------------------------------------|---------------------------|----------------------------|--------------------|--------------------------------|
| UiO                                               | Fox                                 | UiO GPT                   | TSD                        | via project        | via a PI                       |
| UiB                                               | local HPC (UiB IT)                  | Copilot                   | SAFE                       | ?                  | ?                              |
| NTNU                                              | IDUN                                | GPT NTNU                  | HUNT Cloud                 | via supervisor     | support agreement needed       |
| UiT                                               | local HPC group; hosts NRIS systems | ?                         | ?                          | ?                  | ?                              |
| UiA                                               | DGX-2 and DGX H100 (CAIR)           | ?                         | ?                          | ?                  | ?                              |
| INN · HiØ · HVL · OsloMet · Nord · Kristiania ... | ?                                   | ?                         | ?                          | ?                  | ?                              |
| NMH · KHiO · AHO                                  | none known                          | ?                         | none                       | no                 | ?                              |
| National Library                                  | DH-lab                              | ?                         | ?                          | n/a                | ?                              |
| Simula · SINTEF · IFE                             | eX3 and own                         | ?                         | ?                          | n/a                | ?                              |





# Inference is not training

## Serving a model

- always on, answers in milliseconds
- a web endpoint that a class, a stage or a museum connects to
- a modest GPU, or many of them, or a laptop
- needs networking, logins and a security review

## Training and fine-tuning

- bursts: hours to weeks, then nothing
- no endpoint, no users, a queue
- the biggest GPUs and the fastest storage
- needs scheduling, checkpoints, data next to compute





# Workflows do not travel

## Laptop to Fox

Different modules, versions, paths. Days of setup.

## Fox to Olivia

Same scheduler, different drivers, containers and rules.

## Olivia to LUMI

NVIDIA to AMD, CUDA to ROCm. Often a rewrite.

## What helps

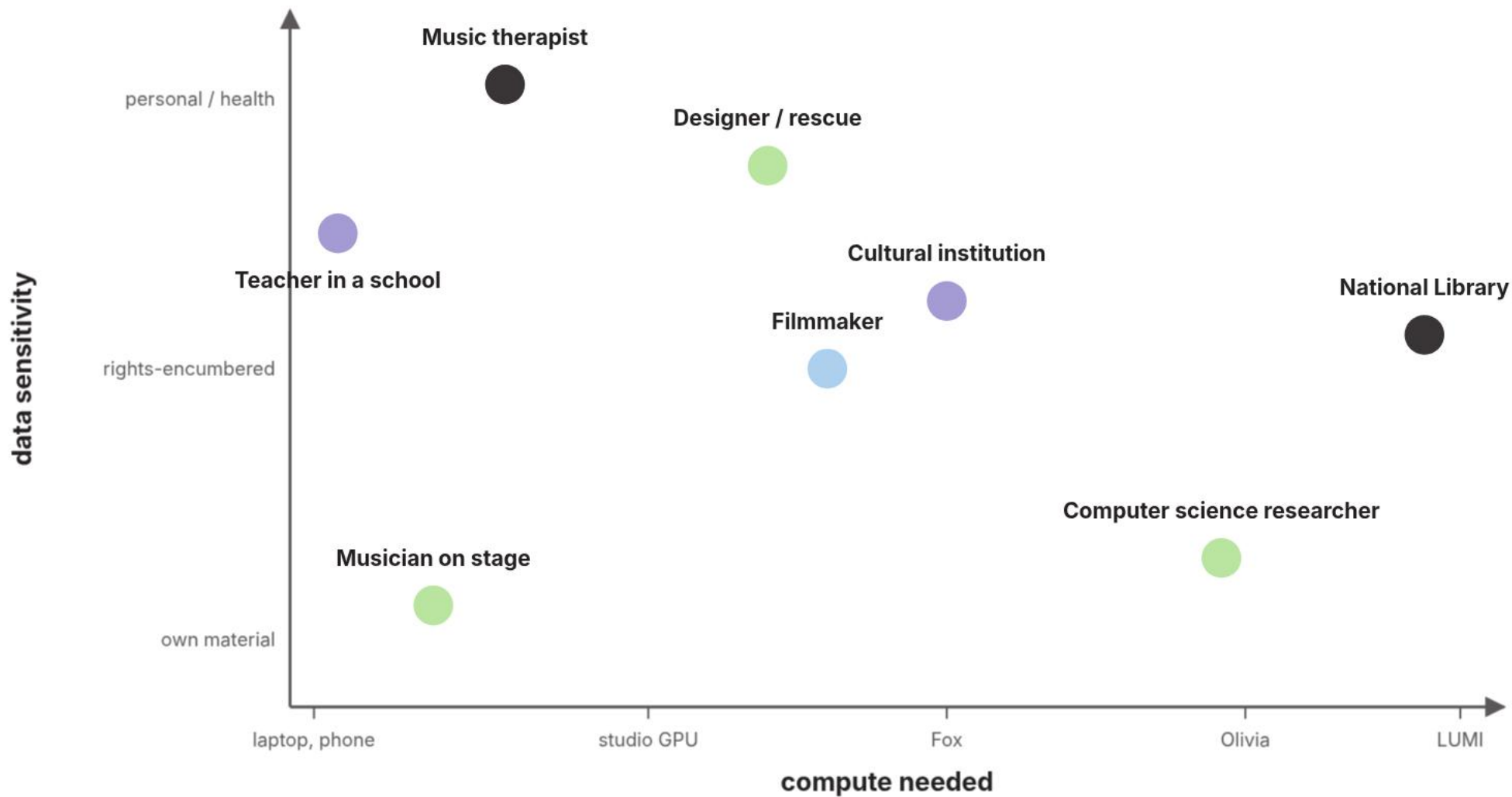
Containers and pinned environments. Teach it at student level. AI assistants make porting cheap.



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MITSH

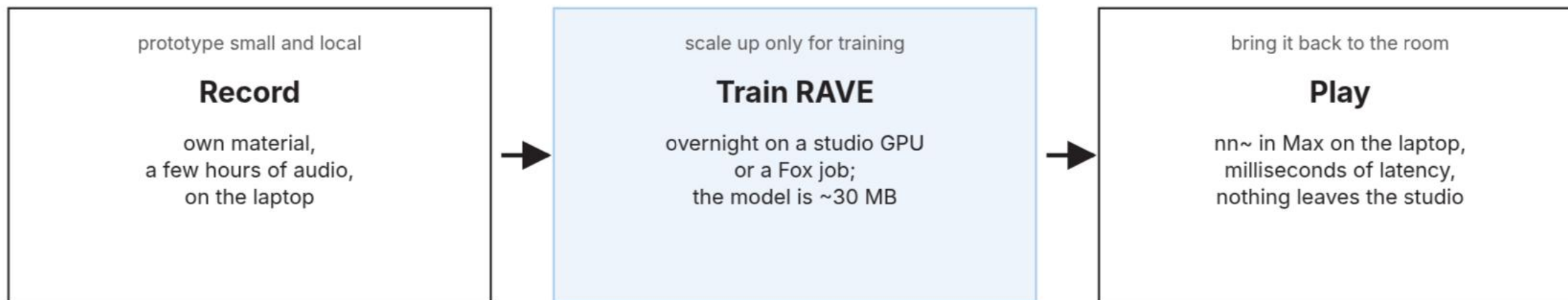
# Who needs what





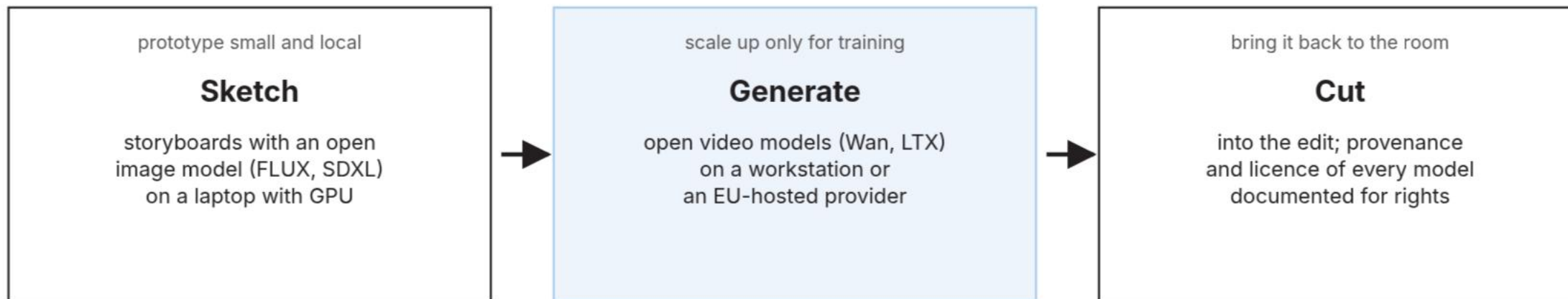


## WP1 · A musician on stage





## WP2 · A filmmaker





## WP4 · A teacher in a school



## WP6 · The National Library

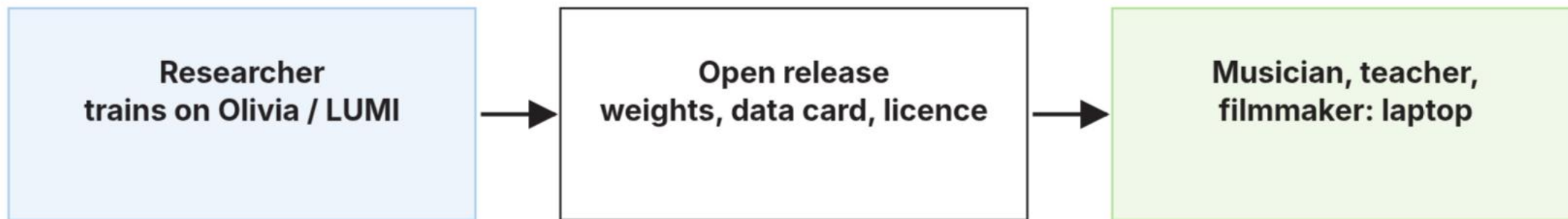




# Work Package Needs

| WP                | Typical machine task                                | Data                              | Compute                             | Runs today on                           | The hole                                                                    |
|-------------------|-----------------------------------------------------|-----------------------------------|-------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------|
| 1 Performances    | real-time co-creative agents, RAVE, gesture models  | own recordings, sensors           | ms; laptop, board, one GPU to train | laptop, studio GPU, Fox                 | open real-time models; training time without paperwork                      |
| 2 Processes       | image, video, text, audio generation                | rights-encumbered                 | hours on a workstation              | commercial tools, workstation           | rights-clean open models; EU hosting                                        |
| 3 Health          | small models on sessions and signals                | health, personal                  | small, inside a secure zone         | TSD, offline laptops                    | GPUs inside secure zones; consent-ready pipelines                           |
| 4 Education       | AI literacy, classroom tools                        | pupils' data                      | none locally; hosted                | tablets and laptops, Feide, hosted chat | free, Norwegian, GDPR-safe creative tools                                   |
| 5 Industries      | production generation, attribution, audits          | contracts, identity               | hosted, reliable, auditable         | commercial APIs                         | cultural institutions in KI-fabrikken; open-provider contracts              |
| 6 Heritage        | transcribe, classify, link, train on the collection | petabytes; copyright, public duty | national or European HPC            | NB DH-lab, Olivia, LUMI                 | multimodal Norwegian models; sustained allocations; storage next to compute |
| 7 Problem-solving | constrained generation, embodied systems            | industrial, safety-critical       | edge to workstation; simulation     | partner systems, Fox                    | constraint-aware models; access for industry partners                       |





**when the loop works: compute upstream, open models downstream**





## Develop it yourself

two different jobs

### Use off the shelf

finished model



any device

hardware: anything that runs inference  
who: everyone on the map

time  minutes

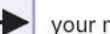
the catch: their data, their terms

### Train an existing architecture on new data

your data



fixed architecture



your model

RAVE on your recordings, a LoRA on your film  
hardware: one GPU, sometimes a cluster  
who: PhDs, labs, a studio with a GPU

time  hours to days, known

the catch: rights to the data

### Develop new methods



new architectures, objectives, control  
hardware: many runs, most fail  
who: ML groups on national compute

time  ?

the catch: allocations want a number first



## CREATIVE AI IS NOT "TRADITIONAL" AI

### **AI for a task**

- one task, one right answer
- batch: throughput matters, latency does not
- large, clean, labelled data
- measured by accuracy on a benchmark
- runs in a data centre, out of sight

### **AI for creative work**

- open-ended: many good answers, or none
- real-time and interactive: latency is the constraint
- small, personal, messy data, often your own
- measured by taste, context and who is asking
- on stage, in the room, on the body: control matters



# Tasks against platforms: a rough guide

| Task                                     | Commercial API | Laptop    | Laptop + GPU  | Workstation | Fox-type cluster | Olivia / LUMI |
|------------------------------------------|----------------|-----------|---------------|-------------|------------------|---------------|
| Chat, writing, code                      | •              | • 7-14B   | •             | • 70B       | 🕒 overkill       | 🕒 overkill    |
| Transcription (Whisper)                  | •              | •         | •             | •           | 🕒 batch          | 🕒 overkill    |
| Image generation                         | •              | 🕒 slow    | •             | •           | 🕒 batch          | 🕒 overkill    |
| Video generation                         | •              | ○         | 🕒 short clips | •           | 🕒 batch          | 🕒             |
| Real-time audio on stage                 | ○ latency      | •         | •             | •           | ○                | ○             |
| Train a small model (RAVE, classifier)   | ○              | 🕒 slow    | •             | •           | •                | 🕒 overkill    |
| Fine-tune an LLM (7-14B)                 | 🕒 data leave   | ○         | 🕒 LoRA        | •           | •                | •             |
| Train from scratch (NB-scale multimodal) | ○              | ○         | ○             | ○           | 🕒 small          | •             |
| Sensitive data (health, pupils)          | ○              | • offline | • offline     | • offline   | 🕒 TSD-type only  | ○             |



# By person: a plausible stack today

| Who                         | Everyday                | Creative work                               | Training                                   | The catch                                    |
|-----------------------------|-------------------------|---------------------------------------------|--------------------------------------------|----------------------------------------------|
| <b>Musician on stage</b>    | laptop, local chat      | Max + nn~, RAVE                             | studio GPU overnight, or a friend with Fox | no institution: no Fox, no NRIS              |
| <b>Filmmaker</b>            | commercial tools        | ComfyUI, open video models on a workstation | LoRA on own footage                        | rights of the base model                     |
| <b>Teacher in a school</b>  | Feide-gated hosted chat | browser tools only                          | none                                       | no Norwegian, GDPR-safe creative tool exists |
| <b>Music therapist</b>      | offline laptop          | small local models                          | inside TSD, if GPUs                        | consent and secure-zone capacity             |
| <b>Bachelor student</b>     | Ollama + Open WebUI     | Whisper, notebooks                          | Colab-type, or a course allocation         | depends entirely on the institution          |
| <b>PhD / ML researcher</b>  | laptop                  | prototypes locally                          | Fox, then NRIS, then LUMI                  | allocation rounds and the paperwork          |
| <b>Cultural institution</b> | commercial APIs         | hosted open models with a contract          | KI-fabrikken, if they qualify              | nobody has defined them as a user category   |
| <b>Startup / studio</b>     | commercial APIs         | own GPUs or cloud                           | KI-fabrikken as "business"                 | cost, and no research partner                |
| <b>National Library</b>     | own lab                 | own models                                  | Olivia, LUMI                               | sustained multi-year allocation              |





# Holes in the map





| Who                                                                                                              | Own hardware             | Institutional<br>servers, VDI, Fox-type | National HPC<br>NRIS, Olivia | KI-fabrikken    | Hosted open models<br>EU providers | Commercial APIs      |
|------------------------------------------------------------------------------------------------------------------|--------------------------|-----------------------------------------|------------------------------|-----------------|------------------------------------|----------------------|
| <b>Researchers, big universities</b><br>UiO · UiB · NTNU · UiT                                                   | yes                      | yes, varies                             | by application               | yes             | own budget                         | own budget           |
| <b>Researchers, other universities and colleges</b><br>UiA · INN · HiØ · HVL · NMH · KHIO · AHO · Kristiania ... | yes                      | often little or none                    | by application               | yes             | own budget                         | own budget           |
| <b>Research institutes</b>                                                                                       | yes                      | own labs                                | by application               | yes             | own budget                         | own budget           |
| <b>Students, bachelor to PhD</b>                                                                                 | own machine              | via a supervisor                        | via a project                | ?               | rarely                             | own budget           |
| <b>Teachers and schools</b>                                                                                      | tablets, laptops, no GPU | no                                      | no                           | no              | no                                 | Feide-gated, limited |
| <b>Freelance artists</b>                                                                                         | if affordable            | no                                      | no                           | ?               | own budget                         | own budget           |
| <b>Public sector</b><br>museums · libraries · NRK · municipalities                                               | yes                      | own IT, no GPUs                         | no                           | "public sector" | budget, procurement                | budget, procurement  |
| <b>Private sector</b><br>studios · film · games · startups                                                       | yes                      | own                                     | only with a research partner | "business"      | budget                             | yes                  |



## Data: open by count, closed by volume

### Open, with an API

- Wikidata, KulturNav, the authority files
- half of DigitaltMuseum, Nasjonalmuseet
- Kartverket, SSB, Stortinget, Brønnøysund
- Språkbanken: thousands of hours of speech

### Closed, by the hour and the page

- the library's books, newspapers, radio, TV, web
- NRK, TV 2, NTB, Retriever
- Bokhylla, Filmarkivet
- nothing open for music, film, video, dance, theatre, games, design



# Observations to argue with

- The academic infrastructure is built for batch science, not for a rehearsal room.
- The people who most need cheap, controllable, rights-clean models have the least access.
- Access depends on which institution you sit in, not on what you need.
- Open Norwegian models exist for language and speech, and for nothing creative.
- The whole Norwegian cultural heritage sits in one collection. Nobody has trained a multimodal model on it.
- Nobody has defined "artist", "teacher" or "cultural institution" as a user of national compute.
- The low-threshold national service (NAIC) is paused; its successor is not yet here.
- Workflows are locked to the system they were built on. Fox is not Olivia, and Olivia is not LUMI.
- When big systems are bought, the users are rarely asked first.
- Norway's data is open by count and closed by volume. The library is licensed for reading, not mining.
- No text-and-data-mining exception in law. The bill is in committee; input is due 28 September.
- 200 researchers on a top-tier subscription is 6 million kroner a year. Per seat is the wrong shape; per use through a gateway is the right one.
- There will never be unlimited access to the best models. Good enough, on a quota, is the plan.
- One PhD fellowship is 150 seat-years. Tokens multiply people; they do not replace them.

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The bill





ONE SUBSCRIPTION · 200 RESEARCHERS

# 6 million kroner a year

200 researchers on one top-tier agent subscription at 2 500 kroner a month.



# Two ways out, and the elephant

## National: quotas through one door

- the gateways and clusters of the big institutions, opened to everyone else on the map
- a quota per person, paid per use by the unit, across vendors
- frontier models when they are needed, open models for the rest
- quota categories for artists, schools and museums, not only researchers

## Local: small and smart

- small open models on the laptop, the board, the phone
- several small agents, one job each, instead of one large model for everything
- energy and cost in the room, where you can see them
- for the stage, the classroom, the museum

The elephant: the bill that matters is not 200 researchers. It is generated feeds and the rest of the consumption nobody asked for. I would argue a centre for AI and creativity has to confront that, and show what else is possible.





# Tokens or people

| What                  | Kroner     | The same money as               |
|-----------------------|------------|---------------------------------|
| One seat, one year    | 30 000     | 1 seat-year                     |
| One PhD year          | 1 400 000  | 47 seat-years                   |
| 200 seats, one year   | 6 000 000  | 1.3 fellowships, or 4 PhD years |
| 200 seats, five years | 30 000 000 | 7 fellowships                   |





# Pay per use, not per seat

## One door

- an institution-hosted gateway in front of commercial and open models: UiO GPT, GPT NTNU
- chat, shared assistants, files, and now API keys for agents and coding tools
- about 50 public-sector institutions run it

## Keys by data class

- one personal key per data class: red, yellow, green
- the key is independent of the model
- a monthly quota per model
- a unit authorises a quota and pays for actual use

## Good enough, not best

- the best open-weight models, served on Fox and IDUN, cover most jobs
- smaller models win on price per result
- the frontier models on a quota when they are needed, without a seat per person



A PROPOSAL TO ARGUE WITH

# How a centre could decide

## A share, not a line

- budget tokens as a share of personnel cost, as equipment always was
- 3 to 5 per cent of a PhD year is 40 000 to 70 000 kroner
- enough for heavy pay-per-use through a gateway

## Quotas by role

- a PhD in machine learning, a WP leader and a museum partner do not need the same budget
- the gateway model, applied to the centre's own money
- the unit sees what its people actually use

## Not fungible

- fellowships are in the contract; tokens come from operating money
- so the real trade-off is tokens against a lab engineer, workshops, travel, hardware in the room
- 50 seats is one engineer who builds the local alternative

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# Over to the panel



- 
- 1 Which tools *can* we use?
  - 2 Which tools *should* we (not) use?
  - 3 Which tools do we need to *develop*?

**Stefano Fasciani**

University of Oslo,  
Department of Musicology

**Enrique Encinas**

Oslo School of Architecture  
and Design

**Anna-Maria Christodoulou**

University of Oslo, RITMO

**Keith Mellingen**

VRINN, immersive learning  
cluster, Hamar



# What do you use, and what can you not get?

[nettskjema.no/a/649352](https://nettskjema.no/a/649352)



Utgiftsrefusjoner

Søk

🔍

(extra) Ai materials for 70409  
CreativePracticeGenerativeAI

Tirsdag 25. august 2026

Til utbetaling

Beløp før skatt: 31 503,35 NOK

Expenses for course 70601 lxD2

Tirsdag 24. februar 2026

Utbetalt

Beløp før skatt: 8 338,66 NOK

Ai materials for 70409 Creative  
Practice with GenerativeAI

Tirsdag 24. februar 2026

Utbetalt

Beløp før skatt: 10 469,87 NOK

Utgiftsrefusjon

Tidsrom

Dato:

25.08.2026

Formål:

(extra) Ai materials for 70409 CreativePracticeGenerativeAI

code the duty

+

keep the beauty

<